

University of Armed Forces “ESPE”

Project's Name: SaborCode

Team's Name: codex

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Problem Definition

Restaurant software project: digital menu, QR code and ordering

1. Project Overview

The project consists of designing and building a web-based software solution for a restaurant. Diners will scan a QR code placed on each table to access an always up-to-date digital menu, and in later phases will be able to order, call a waiter and pay from their own phone. Staff will manage the menu and incoming orders from an administration panel and a kitchen view.

2. Problem Statement

The restaurant depends on printed menus and manual order-taking. This makes menu updates slow and costly, leaves diners with outdated or incomplete information (prices, availability, allergens), and creates avoidable delays and errors between the table, the waiter and the kitchen.

The problem affects diners, waiters, kitchen staff and the owner. Its impact is a slower and less pleasant dining experience, wasted staff time, lost sales from unavailable or poorly presented dishes, and recurring printing costs.

A successful solution would give every diner an accurate, mobile-friendly menu in seconds, let staff update it without technical help, and, in later phases, streamline the path from order to kitchen to table.

3. Background and Operational Context

To understand the environment in which the restaurant operates and the origin of the current problem, the following operational aspects are considered:

- **Operation:** The restaurant operates under a dine-in service model, where the current workflow relies entirely on in-person service provided by the waitstaff.
- **Existing systems:** Currently, the establishment handles order-taking manually (using paper tickets/slips) and presents prices and dishes physically without any digital integration.
- **Connectivity and Access:** The service dynamic depends on customers being able to access menu information from their own mobile devices within the venue.

- **Legal and Regulatory Framework:** The restaurant's commercial operations are subject to local regulations regarding consumer information, allergen disclosures, and sales receipt issuance.

4. Performance Measures

To evaluate restaurant efficiency, order management, and customer experience, the system will compute the following measures:

- **Menu Access Rate:** The frequency and percentage of diners accessing the digital menu via QR codes across different tables and peak hours.
- **Order Processing Time:** A measure of the time elapsed from when a customer places an order until it is confirmed by the kitchen and delivered to the table.
- **Item Popularity and Availability:** Tracking order volumes per dish to identify top-selling items and manage real-time availability.

These measures will help restaurant managers identify service bottlenecks, optimize kitchen workflows, and update menu offerings based on operational data.